

Part 1 insights report on westbound I-94 traffic

This report is for the Smart City Mobility Analytics Team. It uses 48,204 records, representing 40,575 unique hourly observations of westbound I-94 traffic near Minneapolis-St Paul, from 2012-10-02 to 2018-09-30.. Tasks 1 to 3 analysed the records in SQLite, and Task 4 analysed them in Power BI. Every figure below comes from the task notes in this folder.

Key findings

1. Hour of day matters most. In Task 2, hour of day alone explained 78.2% of the variation in hourly volume. In Task 4, average traffic in 2017 was 385.8 vehicles per hour at 03:00, 4,819.5 at 07:00, and 5,834.4 at the 16:00 peak. Task 3 used the brief's definition of congestion, more than 5,500 vehicles in an hour, and 15.05% of hours reached that level.

2. Weather and temperature matter little at a fixed hour of day. Task 2 found that warmer hours have more traffic, but the correlation is weak, at $r = 0.139$. Within the same month and hour of day, r falls to 0.035. In Task 3, clear hours had lower odds of congestion than cloudy hours, with an odds ratio of 0.736. At the same hour of day the odds ratio is 1.006, so the odds are equal. The overall difference appears because clear skies are more common at night, when traffic is light. In Task 4, among weather conditions with at least 900 hours, average traffic was highest under clouds at 3,616.9 vehicles per hour and lowest in fog at 2,703.7. These gaps also depend on the time of day at which each condition occurs.

3. Holidays cut traffic by about a third, whatever the temperature. Task 1.3 compared each holiday with ordinary days of the same weekday in the same month. Labor Day traffic was 65.3% to 67.7% of an ordinary September Monday in 2015, 2016, and 2017. The share stayed in that range although 2017 was 3.7 °C cooler than 2016. On New Year's Day the share depended on the weekday. It was 50.2% on Friday 1 January 2016, 90.0% on Sunday 1 January 2017, and 64.5% on Monday 2 January 2017, the observed holiday.

4. Traffic grew in 2017, but yearly totals do not show it. Data coverage varies substantially between years, ranging from only 23.9% of calendar hours in 2012 to 99.5% in 2017. Consequently, changes in annual totals partly reflect differences in data coverage rather than changes in traffic itself. For example, the original Task 1.2 calculation using the dataset rows showed a 107.99% increase in total recorded traffic from 2015 to 2016. However, when comparable periods with at least 90% coverage were examined, average traffic actually fell by 7.0% from July to September 2015–2016. In contrast, over the comparable May-to-December period, average traffic increased from 3,161.5 vehicles per hour in 2016 to 3,386.7 in 2017, a 7.1% increase. These results show why traffic trends should be assessed using comparable, well-covered periods rather than raw annual totals.

Data quality the team should know about

- The file has no rows between 2014-08-08 and 2015-06-11, a gap of 308 days. In total, 11,976 hours are missing.
- The file repeats an hour once for each weather condition it reports, so 48,204 rows describe 40,575 hours. Summing rows instead of hours overstates yearly totals by 6.9% to 21.1%.
- The holiday column marks only the 00:00 row of each holiday. New Year's Day 2015 has no data.
- 10 temperature readings of 0 K and one hourly rainfall reading of 9,831.3 mm are physically impossible. Two hours on 2016-07-23 record zero vehicles, which is possible but unusual for an interstate.

Implications for mobility decisions

- **Plan around the weekday peaks.** Time incident response and traveller messages to the 07:00 and 16:00 peaks. Hour of day explains 78.2% of the variation in volume, while the correlation with temperature is 0.139.
- **Promote travel just outside the peaks.** In 2017, traffic at 19:00 averaged 3,421.5 vehicles per hour, 41% below the 16:00 peak. Each trip moved from 16:00 to 19:00 lowers the volume in the busiest hour of the day.
- **Use day type in every forecast.** Weekdays, weekends, and holidays have different hourly patterns. Labor Day stayed between 65.3% and 67.7% of an ordinary Monday in all three years. Part 3 models should start from time and day-type features, and they should test weather effects before relying on them.
- **Report traffic per recorded hour instead of raw totals.** A dashboard or trend report should count each hour once and show data coverage alongside traffic volume. Otherwise, changes in recorded totals may be mistaken for changes in underlying traffic, as illustrated by the 107.99% increase in the recorded 2016 total.
- **Fix the data feed before using it for operations.** The 308-day gap, the repeated hours, and the impossible readings suggest faults in data collection. An automated alert built on this feed would react to those faults as if they were traffic events.

Limitations

The data covers one direction of one corridor, so it cannot show where traffic goes when it leaves I-94. The analysis uses observational data, so it shows associations and not causes. The 5,500-vehicle congestion threshold comes from the brief, and the data has no measured road capacity to check it against.